

Joint Models for Marker and Failure Data under Dual Censoring Schemes

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2026 ENAR Spring Meeting, Indianapolis, IN
March 17, 2026



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Motivation

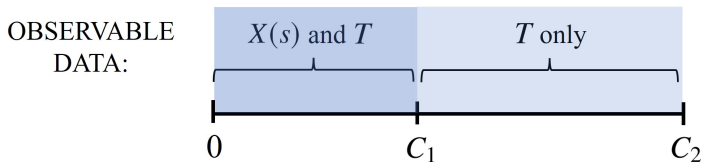
Dual Censoring

Dual censoring occurs when life history processes are censored at two different times.

Boruvka and Cook (2016)

This is common in studies involving longitudinal and failure processes:

- Longitudinal (“marker”) data $\{X(s), s > 0\}$ collected over $(0, C_1]$
- Failure data T collected over $(0, C_2]$, $C_2 > C_1$



$\Rightarrow$ missing marker data over $(C_1, C_2]$

RCT on prostate cancer metastasis to bone:

Saad et al. (2004)

- A biochemical marker of excess bone formation, bone-specific alkaline phosphatase, is measured **every 3 months for 2 years**
- Skeletal-related events and death are recorded **up to 4 years**

Finnish smoking study:

Gutiérrez-Torres et al. (2024)

- Smoking behaviour is recorded **every 4 months for 8 years**
- Lung cancer diagnoses are identified from cancer registry **up to 27 years**

Past Work – Multiple Imputation

One option for accommodating the missing marker data is *multiple imputation (MI)*.

MI for failure time models has been investigated:

- Some methods yield estimates that are only **approximately valid** (even asymptotically), which may lead to bias White and Royston (2009); Murad et al. (2020)
- Other methods have been developed, but imputing **time-varying covariates** is still challenging Bartlett and Taylor (2016); Bonneville et al. (2025)
 - Imputation of $X(s)$ involves a model for $X(s) \mid X(s+1), X(s+2), \dots$

Extending the method of Bartlett and Taylor (2016) to time-varying covariates could be an interesting line of investigation.

Past Work – Joint Modeling

Another option is to *jointly model* the marker and failure processes.

Types of joint models:

- Shared/correlated random effect models Rizopoulos (2012)
 - R software: `joiner`, `JM`, `JMbayes`, `frailtypack`, etc.
- Joint latent class models Proust-Lima et al. (2014)
 - R software: `lcmm`
- Multivariate copula joint models Zhang et al. (2026)
- Multistate models Cook and Lawless (2018)
 - R software: `survival`, `msm`

Multistate models tend to be robust for estimates of state occupancy Aalen et al. (2001), but these usually require the **same length of follow-up** for the marker and failure processes.

Goal: Investigate strategies for studying the relationship between a marker $X(s)$ and failure time T under a dual censoring scheme.

Approach: Formulate a joint multistate model for $X(s)$, T , and the random censoring process, then use an EM algorithm to accommodate the missing marker data between C_1 and C_2 .

To focus on the dual censoring issue, we consider a simplified setting:

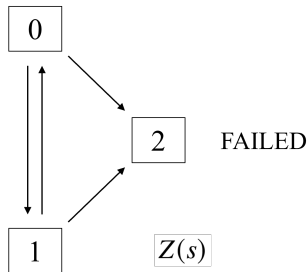
- Discrete time
- Binary marker $X(s)$
- Failure time T and random censoring times C_1, C_2 all mutually independent

Joint Model for Marker, Failure, and Random Censoring Processes

Notation

Marker-failure process:

NORMAL
MARKER



ELEVATED
MARKER

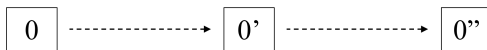
- $N(s) = 1$ if $T \leq s$ (failed by time s)
0 otherwise
- $X(s) = 1$ if marker is elevated and $N(s) = 0$
0 if marker is normal and $N(s) = 0$

Random censoring process:

UNCENSORED

CENSORED
FOR MARKER

FULLY
CENSORED

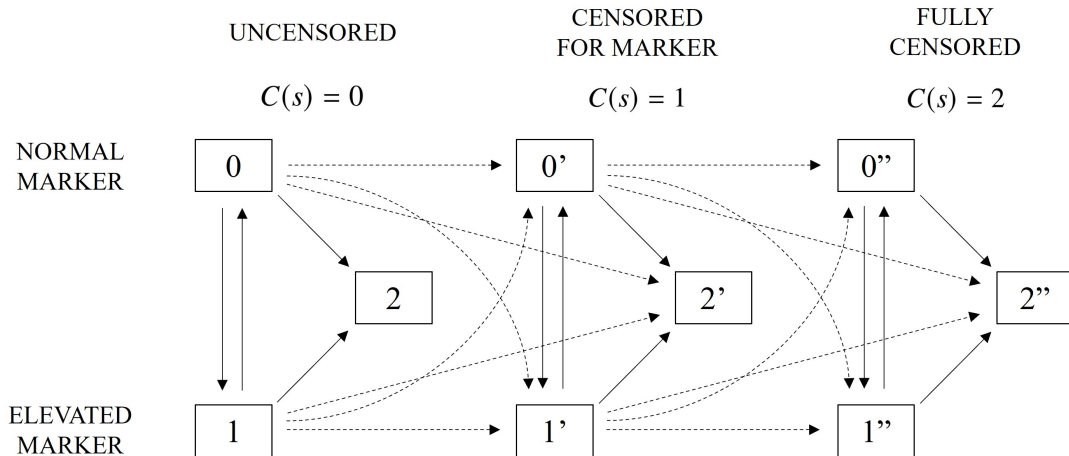


$C(s) = 0$

$C(s) = 1$

$C(s) = 2$

Joint Multistate Diagram



Discrete-Time Intensity Models

Let V_i be a time-fixed binary covariate. For $z_i, v_i \in \{0, 1\}$:

Intensity for failure:

$$\begin{aligned}\mathbb{P}(Z_i(s) = 2 \mid Z_i(s-1) = z_i, C_i(s-1), V_i = v_i) \\ = \text{expit} \{ \beta_0(s) + \beta_1 z_i + \beta_2 v_i + \beta_3 z_i v_i \} = \pi_i^N(s; \beta)\end{aligned}\quad (1)$$

Intensity for elevated marker:

$$\begin{aligned}\mathbb{P}(Z_i(s) = 1 \mid Z_i(s) \neq 2, Z_i(s-1) = z_i, C_i(s-1), V_i = v_i) \\ = \text{expit} \{ \alpha_0(s) + \alpha_1(s) z_i + \alpha_2 v_i + \alpha_3 z_i v_i \} = \pi_i^X(s; \alpha)\end{aligned}\quad (2)$$

Random censoring intensities: (independent and non-informative)

$$\begin{aligned}\mathbb{P}(C_i(s) = 1 \mid Z_i(s-1), C_i(s-1) = 0, V_i = v_i) &= \text{expit}(\gamma_1) \\ \mathbb{P}(C_i(s) = 2 \mid Z_i(s-1), C_i(s-1) = 1, V_i = v_i) &= \text{expit}(\gamma_2)\end{aligned}\quad (3)$$

Likelihood Construction

Let τ be an administrative censoring time.

The **complete data** likelihood corresponds to **no early marker censoring**:

$$\mathcal{L}_i(\alpha, \beta) \propto \prod_{s=1}^{\tau} \left\{ \pi_i^N(s; \beta)^{\mathbb{I}(Z_i(s)=2)} [1 - \pi_i^N(s; \beta)]^{\mathbb{I}(Z_i(s) \neq 2)} \right. \\ \left. \cdot \pi_i^X(s; \alpha)^{\mathbb{I}(Z_i(s)=1)} [1 - \pi_i^X(s; \alpha)]^{\mathbb{I}(Z_i(s)=0)} \right\}^{\mathbb{I}(Z_i(s-1) \neq 2)} \quad (4)$$

In the **observed data**, we do not know $Z_i(s) = 0$ vs. 1 when $C_i(s) = 1$.

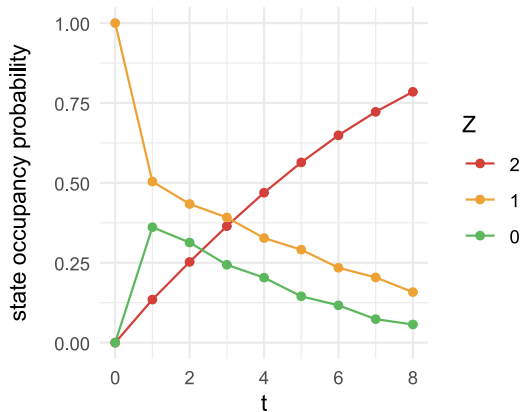
We use an EM algorithm:

Dempster et al. (1977)

- Weighted logistic regression with a pseudo-subject for each possible sequence of $Z_i(s) \in \{0, 1\}$, each weighted by its conditional probability given the observed data
- Variance estimation via Louis (1982)

Efficiency with a Correctly Specified Joint Model

Parameter Settings – Example



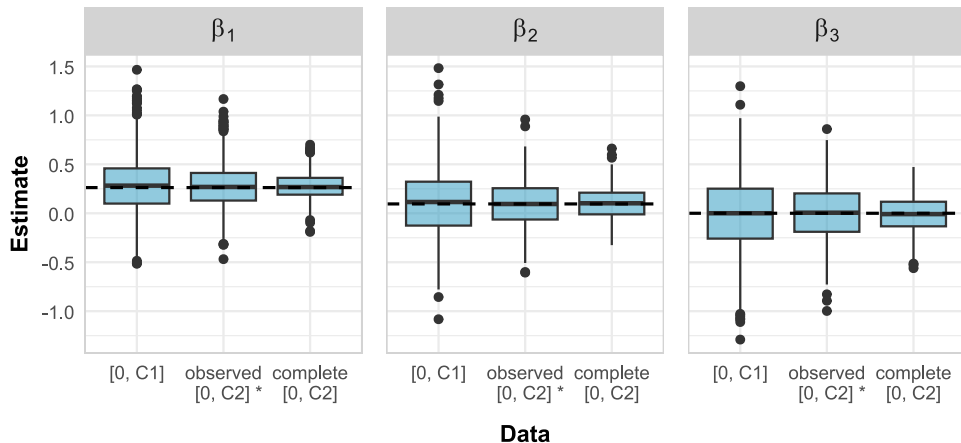
Administrative censoring at $\tau = 8$.

% expected marker data missing:

$$\begin{aligned}\mu &= \frac{\mathbb{E}\{\# \text{ visits with } C(s) = 1\}}{\mathbb{E}\{\# \text{ visits with } C(s) \in \{0, 1\}\}} \\ &= \frac{\sum_{s=1}^{\tau} \mathbb{P}(C(s) = 1)}{\sum_{s=1}^{\tau} \mathbb{P}(C(s) < 2)} = 0.4\end{aligned}$$

Simulation Results

FAILURE PROCESS PARAMETERS



* uses proposed method

Sensitivity to Misspecification of the Marker Model

True & Fitted Models

We now consider misspecification of the marker model

$$\mathbb{P}(Z_i(s) = 1 \mid Z_i(s) \neq 2, Z_i(s-1) = z_i, V_i = v_i) = \pi_i^X(s; \alpha)$$

Time non-homogeneous process:

True model: $\pi_i^X(s; \alpha) = \text{expit} \{ \alpha_0(s) + \alpha_1(s)z_i + \alpha_2v_i + \alpha_3z_iv_i \}$

Fitted model: $\pi_i^X(s; \alpha) = \text{expit} \{ \alpha_0 + \alpha_1z_i + \alpha_2v_i + \alpha_3z_iv_i \}$

Omitted covariate:

True: $\pi_i^X(s; \alpha) = \text{expit} \{ \alpha_0(s) + \alpha_1(s)z_i + \alpha_2v_i + \alpha_3z_iv_i + \alpha_4v_i^* + \alpha_5z_iv_i^* \}$

Fitted: $\pi_i^X(s; \alpha) = \text{expit} \{ \alpha_0(s) + \alpha_1(s)z_i + \alpha_2v_i + \alpha_3z_iv_i \}$

Heterogeneous Markov processes:

True: $\pi_i^X(s \mid u_i; \alpha) = 1 - \exp \{ - \exp \{ \alpha_0 + \alpha_2v_i + u_i \} \}$ for random effect u_i

Fitted: $\pi_i^X(s; \alpha) = \text{expit} \{ \alpha_0(s) + \alpha_1(s)z_i + \alpha_2v_i + \alpha_3z_iv_i \}$

Effects of Misspecification on Inference

Let $\tilde{\vartheta}$ denote the estimator of $\theta = (\alpha', \beta')'$ under a given misspecified fitted model. Then

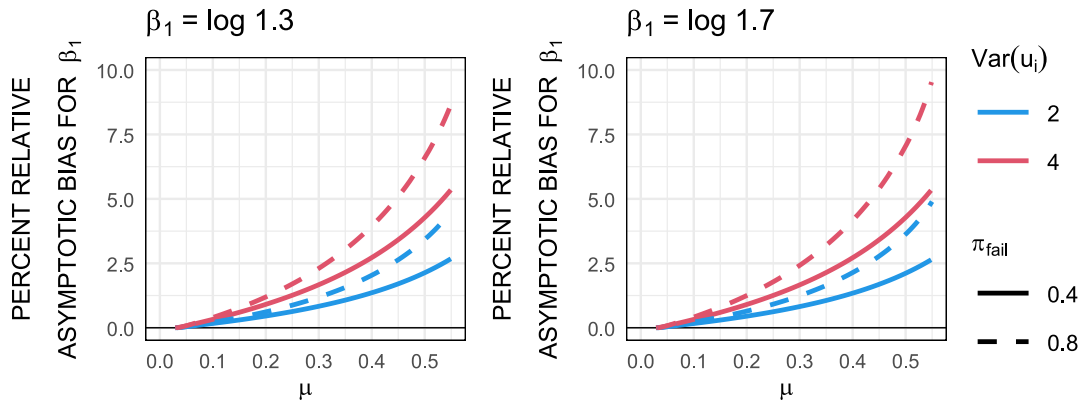
$$\sqrt{n}(\tilde{\vartheta} - \vartheta^*) \xrightarrow{D} MVN(0, \Sigma) \quad \text{White (1982)}$$

where Σ is calculated from the misspecified score function using a sandwich form.

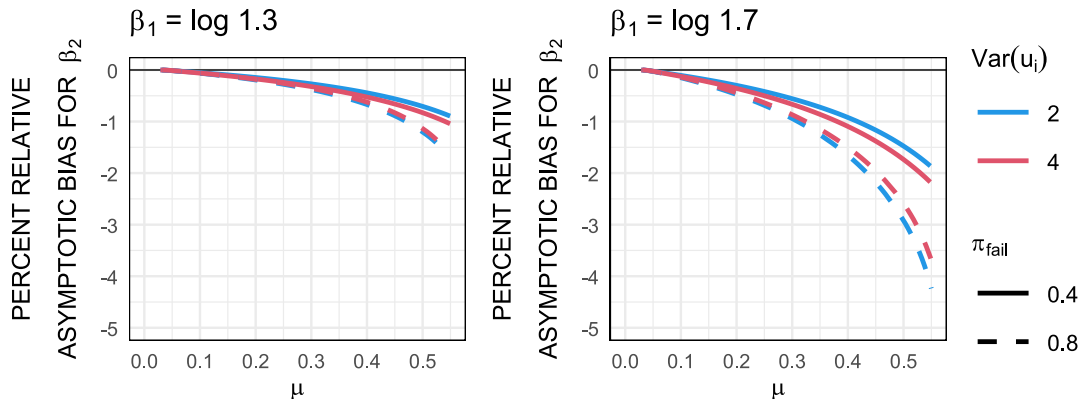
We calculated ϑ^* and Σ for a variety of parameter settings in each of the three scenarios.

The following slides show example results from the *heterogeneous Markov process* scenario, for which the **magnitude of the percent relative asymptotic bias** was the greatest.

Asymptotic Results – Heterogeneous Markov Process



Asymptotic Results – Heterogeneous Markov Process



Discussion

Key takeaways:

- Using the proposed multistate approach with a correctly specified model improves efficiency substantially when estimating parameters for the failure process
- Estimation and inference about the failure process are quite robust to the marker model misspecifications considered here

Extensions in progress & future work:

- Continuous time
- More complicated latent multistate disease processes (e.g. add competing risks)
- More complicated observation processes (e.g. intermittent observations, different censoring times for competing risks)

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Thank you!



`lbumbulis.github.io`